DATS 4001

Detection of Impersonated Accounts on Instagram

Phuong-Anh Pham

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Problem

- With 10 million Instagram profiles that are impersonated cause danger to individual and company safety.
  ➔ In 2025, 260 scammers were arrested for stealing $2.8 million from over 1,400 victims through romance and extortion scams.
  ➔ Scammers use AI to generated profiles for identity theft and blackmail, recently seen in Pune, india, where a fake account was used to harass and extort a colleague.

- **Objectives Question:** What are the most significant indicators or features to find any impersonation account patterns/ account?

- Why does it matter?
  - Reduce scale of threat
  - Financial & safety impact
  - Allowing security or control over users personal data.

The Data

- Source
  - **Web-Scraping:** Collected live account data from Instagram using Selenium and BeautifulSoup.
  - **Third-Party Platforms:** Aggregated influence and follower metrics from Instrack, ViralPitch, and Hopper HQ.
  - **Raw Datasets:** integrated multiple CSV files containing profiles for both real and fake users on GitHub.

- Preparation
  1. **Feature Selection:** Filtered datasets to keep essential columns like name, username, post count, and bio.
  2. **Standardization:** Renamed inconsistent column headers across different sources to a unified format.
  3. **Data Merging**: Combined separate real and fake user files into a single, comprehensive master dataset.
  4. **Data Cleaning:** Removed duplicate entries, handled missing values (NAs), and standardized data types for analysis.

Next Steps

Limitation

- Not enough data
- Mistakenly flagged new users

Future Improvement

- Collaborate with social media company (Meta for Instagram) to increase data.
- Advance research detect impersonated account that is not just account information. (AI picture or chat)

Thank You

Q & A